

4-state EKF controller (gain rate fed forward from the desired trajectory). Built on the 5-state (disturbance state removed). The gain rate δa_x is no longer an estimated state: it is computed from the known desired trajectory through the gain height-sensitivity, $\delta a_x[k] = a'_x[k] \Delta h_d[k]$. The augmented per-axis state shrinks to $\mathbf{x} = [\delta x_1, \delta x_2, \delta x_3, a_x]^\top$. Plant, Measurement, Control objective, Ideal control effort, and the disturbance removal are identical to the 6-state baseline (6state_derivation_xd.tex) / 5-state and are not reprinted; only the gain dynamics and everything downstream change.

Implementable control effort

$$f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k]$$

(4-state: δa_x is no longer estimated — it is the known feedforward below. The rate state is dropped; only the single gain state a_x remains.)

$$\delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k] \Delta h_d[k] \quad (\text{known}), \quad \delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k] \quad (\text{random})$$

where $\Delta h_d[k] = h_d[k+1] - h_d[k]$ is the desired-trajectory height increment and the sensitivity is evaluated on the *desired* trajectory: $a'_x = da_x/dh = -a_{x_{det}} K_h(\bar{h}_d)/R$ with $a_{x_{det}} = a_{nom}/c(\bar{h}_d)$, so $\delta a_x[k] = -a_{x_{det}}[k] K_h(\bar{h}_d[k]) \Delta \bar{h}_d[k]$. The multiplier is the *exogenous* $a_{x_{det}}$, **not** the estimate $\hat{a}_x$ — this keeps the feedforward free of an estimate $\rightarrow Q \rightarrow$ estimate bias loop. The random increment $\delta a_{ram} = a'_x \Delta x_{ram}$ shares the same a'_x .

(Approximation: $a_{x_{det}}$ uses the desired height $\bar{h}_d$, not the true height. The residual $a_{x_{det},true} - a_{nom}/c(\bar{h}_d) \approx a'_x \delta x$ is second order (tracking error $\times$ gain slope); benign away from the wall, but grows near it where $a'_x \propto a_x K_h$ is large.)

Parameter estimator:

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \delta a_x[k] \quad (+ \text{ measurement correction; } \delta a_x \text{ fed forward})$$

Parameter estimation error: $e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$ (no $e_{\delta a_x}$ — the rate is known)

Estimation error dynamics (open loop):

$$e_{a_x}[k+1] = e_{a_x}[k] + \delta a_{ram}[k]$$

(δa_x enters both a_x and $\hat{a}_x \Rightarrow$ it cancels in the error. The gain error no longer carries the deterministic trajectory variation, so it does not lag during fast motion; only the random δa_{ram} drives it.)

Motion error

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \sum_{j=1}^i \delta a_{ram}[k-j]$$

(vs 5-state: the $-i e_{\delta a_x}$ rate-error term is gone, since δa_x is known.)

$$\boxed{\begin{aligned} \delta x[k+1] &= \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x_D^d[k] \\ &\quad - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta af}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]} \end{aligned}}$$

(vs 5-state: the $+dF_{dx} e_{\delta a_x}$ coupling vanishes; ε unchanged. δx_D^d stays uncompensated and is modelled as 0 by the EKF.)

where $F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i]$.

State equation

(EKF model: $\delta x_D^d \equiv 0$; δa_x known feedforward)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = a_{xm}[k] = a_x[k-d] + n_a[k] \end{cases}$$

(4-state: a_{xm} is unchanged. Because δa_x is known, the d -step gain drift goes into the measurement prediction, not into $\mathbf{H}$.)

$$\hat{y}_2[k] = \hat{a}_x[k] - \sum_{i=1}^d \delta a_x[k-i], \quad e_{y_2}[k] = y_2[k] - \hat{y}_2[k] = e_{a_x}[k] + \underbrace{\left(n_a[k] - \sum_{i=1}^d \delta a_{ram}[k-i] \right)}_{r_2[k]}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k], \quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}, \quad e_{y_2}[k] = e_{a_x}[k] + r_2[k]$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \delta a_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = e_{a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \underbrace{\delta a_{ram}[k]}_{q_4[k]}$$

(4-state: no $e_{\delta a_x}$ row, and no $\pm \ell d e_{\delta a_x}$ terms — $\mathbf{H}$'s second row is clean $[0\ 0\ 0\ 1]$.)

$\mathbf{F}_e[k]$ (4×4)

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L} \mathbf{r}[k]$$